

- 2 (a) Inheritance of petal colour and pollen shape in sweet pea plants are controlled by two genes.
- Gene **P/p** controls petal colour. Allele **P** for purple petals is dominant to allele **p** for red petals.
 - Gene **G/g** controls the shape of pollen grains. Allele **G** for long pollen grains is dominant to allele **g** for round pollen grains.

Fig. 2.1 shows the structures inside a flower from a sweet pea plant.

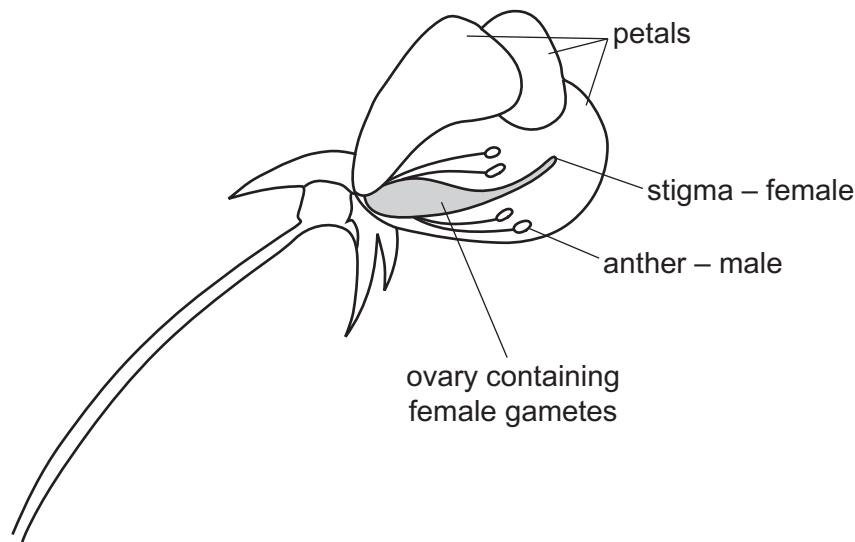


Fig. 2.1

Sweet pea plants are normally **self-fertilising**. Pollen grains containing the male gametes are produced in the anthers and released into the flower. The pollen grains fall on the stigma in the flower, leading to fertilisation of the female gametes.

A plant breeder **cross-fertilised** two sweet pea plants, **A** and **B**.

A homozygous dominant plant with purple petals and long pollen grains (plant **A**) was crossed with a homozygous recessive plant with red petals and round pollen grains (plant **B**).

The breeder transferred pollen grains from the flowers on plant **A** to the flowers on plant **B**.

- (i) Suggest **one** way in which the breeder could prevent the self-fertilisation of plant **B**.

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- (ii) Suggest **one** way in which the breeder could transfer pollen grains from an anther to a stigma.

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(b) The plant breeder then:

- collected the seeds from plant **B**
- grew the seeds in a glasshouse.

The environmental conditions were standardised.

State **three** variables that should have been standardised in the glasshouse.

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[3]

- (c) All the F1 plants grown from the seeds collected from plant **B** had purple petals and long pollen grains.

The plant breeder then carried out a test cross by crossing the F1 plants with homozygous recessive plants with red petals and round pollen grains.

Table 2.1 shows the results of the test cross.

Table 2.1

offspring phenotypes	frequency
purple petals long pollen grains	102
red petals round pollen grains	112
purple petals round pollen grains	14
red petals long pollen grains	20
total	248

The plant breeder expected the test cross to result in a 1:1:1:1 phenotypic ratio.

- (i) Complete Table 2.2 **and** calculate the value of chi-squared (χ^2) for the results of the test cross.

The equation for the calculation of χ^2 is:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

O = observed result

E = expected result

Σ = sum of

Table 2.2

offspring phenotype	O	E	$\frac{(O-E)^2}{E}$
purple petals long pollen grains	102	62	
red petals round pollen grains	112	62	
purple petals round pollen grains	14	62	
red petals long pollen grains	20	62	
$\chi^2 =$			

[2]

The null hypothesis of this chi-squared test (χ^2) is:

There is no difference between the observed and expected results.

The critical value at $p = 0.05$ is **7.82** at 3 degrees of freedom.

- (ii) Use your calculated value of χ^2 to:

- explain whether the null hypothesis should be accepted or rejected

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- suggest a conclusion the plant breeder could make about the inheritance of the genes controlling petal colour and pollen grain shape in sweet pea plants.

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- (d) Some individuals of the plant species *Mimulus guttatus* show copper tolerance as they are able to grow on land heavily polluted with copper.

Copper tolerance is controlled by an allele of a single gene. Plants that do not have this allele are sensitive to high concentrations of copper(II) ions in the soil and show poor growth.

Scientists wanted to investigate if pollen grains are sensitive to copper(II) ions.

Pollen viability is the ability of pollen to produce male gametes that can fertilise a female gamete.

The scientists investigated the viability of pollen grains by placing them into a culture medium. Normally, after several days pollen grains start to grow pollen tubes. Pollen tubes grow down into the ovary to carry the male gametes to the female gametes.

- Pollen grains were collected from copper-sensitive and copper-tolerant plants of *M. guttatus*.
- Pollen grains were placed in culture media as shown in Table 2.3.

Table 2.3

type of plant	number of pollen grains added to culture medium	
	culture medium containing no copper(II) ions	culture medium containing copper(II) ions
copper-tolerant	200	200
copper-sensitive	200	200

All other variables were standardised.

The results are shown in Fig. 2.2.

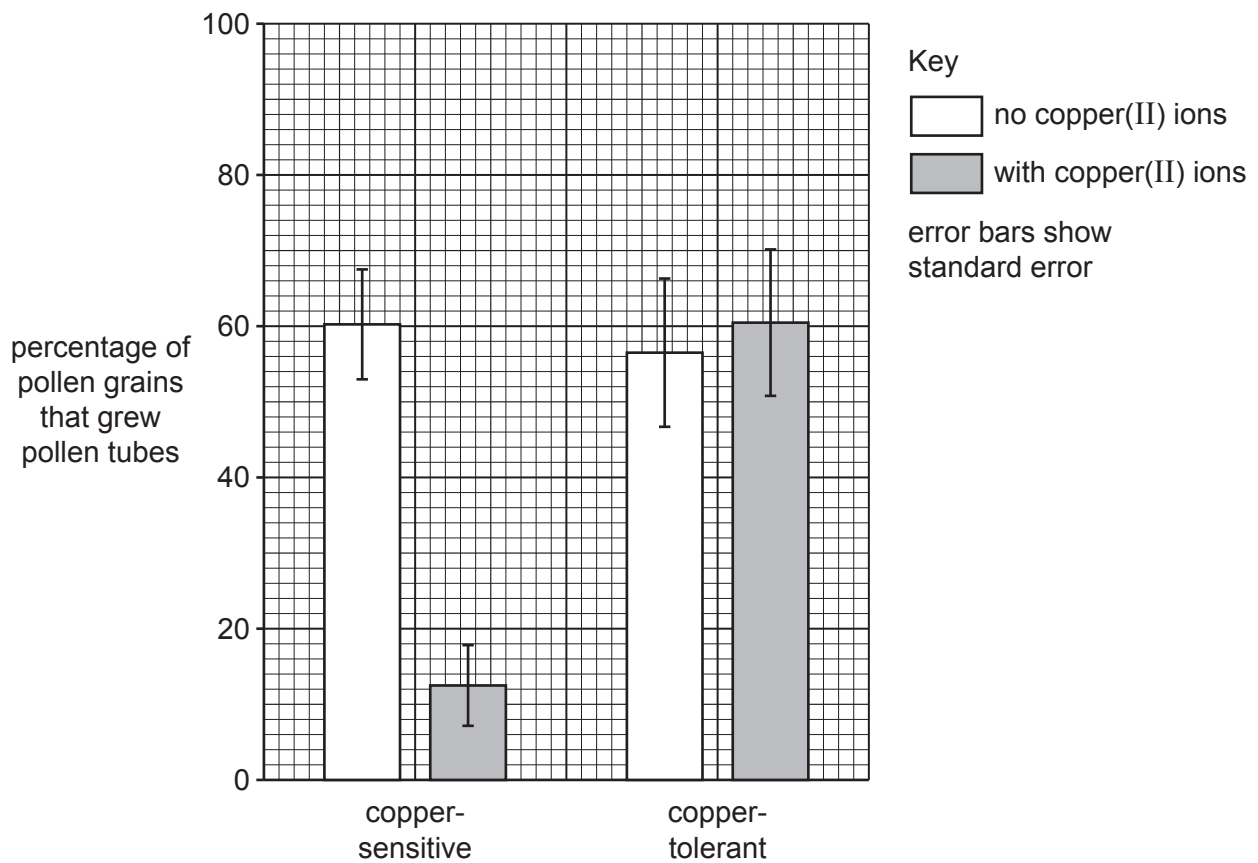


Fig. 2.2

The scientists concluded that the presence of copper(II) ions has a significant effect in reducing the viability of pollen grains.

Explain whether or not the scientists are justified in this conclusion.

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[Total: 12]